Detecting Extrasolar Planets using the Transit Method

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Abstract

The transit method of extrasolar planet detection arises from the phenomenon that, when a planet passes between the Earth and a star, there is a very slight decrease in the recorded brightness of the star. By taking multiple images before and during a suspected transit of CoRoT-1b in front of CoRoT-1, we used a method of differential photometry to determine the flux of the star and observed a noticeable decrease in brightness during the suspected transit. We used literature values to determine characteristics of CoRoT-1, including its radius and spectral class, based on it's absolute magnitude. The stellar radius allowed us to infer the exoplanet's radius, and we made use of Kepler's Third Law and the radial velocity of CoRoT-1 to find the exoplanet's mass. The findings of this experiment were that CoRoT-1b has a radius of $1.682 \times R_{\rm Jup}$ and a mass of $1.41 \times M_{\rm Jup}$. We found these values to be within two standard uncertainties of the literature (Bonomo et al. 2017).

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1 Introduction

Extrasolar planets, or exoplanets, are planets that orbit distant stars outside of our solar system. The first discovery of an extrasolar planet was made in the 1990s. Since then, we have discovered many thousands more. Differential photometry is just one of many different types of aperture photometry that exist. Some others we use are surface photometry (for extended objects like comets or galaxies) and relative photometry (comparing flux of a star to stars with known magnitudes).

To detect exoplanets, we can make use of the transit method and the radial velocity method. These methods use Charge-Coupled Devices (CCDs) which enable precision photometry over the whole sky by capturing photons across regions of the sky. We can make counts of these photons to determine flux values of stellar objects. A transit occurs when an exoplanet passes between us and its star, resulting in a change in magnitude that can be detected and quantified. In conjunction with this, in a two-body system both the star and the planet follow an orbital path. We can measure the Doppler shift that arises from the star's radial velocity which will be important to finding the exoplanet's mass.

To determine CoRoT-1b's radius, we will be measuring the percentage drop in flux over a suspected transit of the planet in front of the star by comparing the flux to another star in the image. We will use literature values of CoRoT-1 to determine the stellar radius, and then use the formula

$$\frac{R^2_{\text{planet}}}{R^2_{\text{star}}} = \%\text{ change in flux.} \tag{1}$$

to find the planet's radius.

Kepler's third law will be important in determining the planet's mass,

$$P^2 = \frac{4\pi^2 r^3}{GM}, \tag{2}$$

where P is the period (s), r is the semi-major axis (m), G is the universal gravitational constant and M is the mass of the central body (kg).

In extension to finding the radius and mass of CoRoT-1b, in this report we will also explore possible areas of uncertainty in our data, comparing our results to well-defined literature values, and exploring exciting areas of research that are being performed using aperture photometry, beyond the transit of a single planet.

2 Method and Results

2.1 Differential Photometry

Twelve images of CoRoT-1 were taken over a period of 50 minutes. They were each taken in the R-band with a 60-second exposure. Between the 6th and

7th image there was a 35-minute gap, intended to capture a suspected transit of CoRoT-1b. Appendix 1, Figure 2, shows the first image taken during this process. The analysis of these images was aided with the use of the program ds9, which allows us to determine total flux counts across regions of the images to determine relative flux values. In Appendix 1, Figure 3, I have included an annulus over CoRoT-1. The purpose of taking an annulus is so that we can use the outer-most region to find a per-pixel background flux value and subtract this from the innermost region. The middle region acts as a buffer to reduce noise from the target star. Our goal is to catch some short decrease in the star's brightness. If there was no transit, the ratio between the brightness of CoRoT-1 and another arbitrary star in the image would remain constant throughout the capture period. This motivates the start of the analysis. By drawing an annulus around the star in each of the 12 images and determining the flux values, as well as doing the same process for another star in the image, we can observe how the ratio between the flux of both stars varies throughout the capture period. I chose to use the bright star immediately down to the right of CoRoT-1 in the image.

Table 1 is a tabulated list of all the relevant flux data we have obtained from all the images.

Image Number	Flux of CoRoT-1	Flux of reference star	Flux ratio
1	120219.7	293925.6	0.4090
2	123072.6	297770.7	0.4133
3	121246.4	297056.0	0.4082
4	117504.0	285082.0	0.4122
5	114791.7	280967.5	0.4086
6	113369.2	269479.9	0.4207
7	115511.5	289977.9	0.3983
8	117260.6	286019.2	0.4100
9	119463.7	294963.3	0.4050
10	110123.7	271422.1	0.4057
11	114313.5	283973.3	0.4026
12	110794.3	275988.7	0.4014

Table 1: List of flux data for every image. The units of flux here are the 'pixel values' from the original image, a unit of intensity. Flux ratios are included. The original spreadsheet of individual data points is found in Appendix 2.

Importantly, we can now make a plot of the flux ratios before and after the suspected transit.

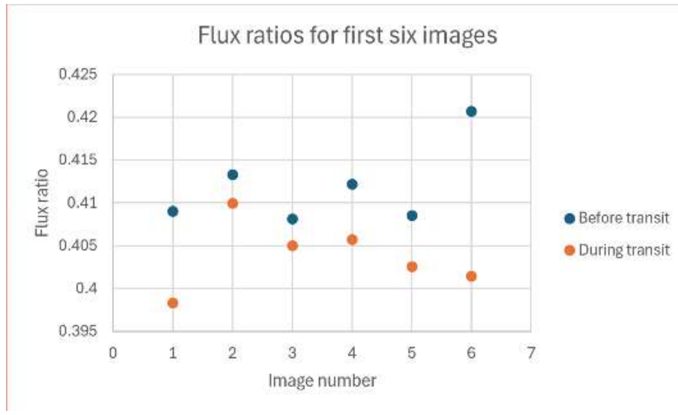

Figure 1: Flux ratios between CoRoT-1 and a reference star plotted against image number, before and during a suspected transit.

To determine whether this result is statistically different, we can determine whether there is an overlap between the mean of each group within three standard errors. The mean value for the first six images is $\bar{x} = 0.4120$, and for the second six images, $\bar{x} = 0.4038$. To take the standard error, we use the standard deviation divided by the square root of the number of measurements in the data group.

$$e = \frac{\sigma}{\sqrt{N}} \tag{3}$$

The standard error for the first group is $e = 0.0018$ and the the second group the standard error is $e = 0.0015$. This reveals that there is **no overlap to within three standard errors of our results**. We can conclude that the two groups are **statistically different**. This is enough information to infer that there was a transit of a planet in front of CoRoT-1, which we are assuming to be CoRoT-1b. To confirm this, we will infer some characteristics of the exoplanet and compare this with literature values.

2.2 Determining the radius

We may first determine that the percentage difference of the flux during the transit is a drop of 1.98%. This is determined by taking the difference in flux ratio and dividing it by the full flux before the transit. The absolute magnitude in the V-band, M_V, is 4.067[1]. The literature table in Appendix 3, Figure 5, shows an extensive list of main-sequence stars classed according to spectral type. Using this lookup table, we can determine that CoRoT-1's absolute magnitude represents an F5 class star with radius of **1.2 solar radii**. The radius of the exoplanet is found using the stellar radius and the percentage change in flux,

[1]Determined by taking the distance modulus, knowing the apparent magnitude $m_V = 13.6$ and $d = 806.45$pc using the distance-parallax formula (GAIA parallax of $1.24 \cdot 10^{-3}$")

using the equation

$$\frac{R^2_{\text{planet}}}{R^2_{\text{star}}} = \%\text{ change in flux.} \tag{4}$$

Using equation 4, we find that $R_{\text{planet}} = 1.1758 \cdot 10^5$km, or equivalently, the radius of CoRoT-1b is $1.682 \times R_{\text{Jup}}$.

2.3 Determining the mass

This process is slightly more abtract than finding the radius. Let's start by stipulating Kepler's third law[2] rearranged for the planet's mass,

$$m_p^3 = \frac{4\pi^2 m_s^2 a_s^3}{GP^2}, \tag{5}$$

where m refers to mass (kg), a refers to semi-major axis (m), G is the universal gravitational constant, and P is the period (s). Our stellar mass using Figure 5 is 1.4 solar masses, and we know that the period is 1.509 days (Monash 2024). Our only remaining unknown is the stellar semi-major axis. The radial velocity diagram of CoRoT-1 is attached in Appendix 4, Figure 6. By examining the plot, we can determine the orbit of CoRoT-1 to be circular as the plot appears to be approximated by a perfect sine function. The rotational velocity is the amplitude of the plot, approximately 0.2km/s. Rotational velocity is equal to circumference divided by period, hence 0.2km/s $= \frac{2\pi a_{\text{star}}}{1.509\text{ days}}$. The stellar semi-major axis is 4150.05km.

Since we now have all the required values, we can return to Equation 5 to determine that the exoplanet mass is $2.68 \cdot 10^{27}$kg, or more appropriately $1.41 \times M_{\text{Jup}}$.

3 Discussion

Some possible sources of uncertainty in our differential photometry recordings would come from both limb darkening and the inclination of the orbit. Limb darkening would affect our results as less flux comes from the edges of the star. Since stars are gaseous and not completely opaque, as you start to record the photons coming from the edges of the star, further away from the centre, there is less hot matter emitting photons. For a potential exoplanet to be in transit, we don't yet have a way of determining where in its transit it is located. We may be observing less of a difference in photons if the exoplanet is nearer to the edges of the star, making a transit harder to detect. In addition, the inclination of the exoplanet's orbit will skew our results. Clearly, if the orbit is skewed so much that the exoplanet does not ever pass in front of the star from our view, then we will never be able to detect it using this method. Conversely, if it passes across the lower or upper portion of the star, we will get a consistently decreased

[2] We will also make use of the conservation of angular momentum which tells us $(ma)_{\text{planet}} = (ma)_{\text{star}}$

change in flux value across the transit. This will yield consistently inaccurate results in our calculations as we have detected a smaller flux difference than would otherwise be detected, had the planet crossed closer to the centre of the star. These are a few considerations to keep in mind when determining the characteristics of the exoplanet.

It is also important to compare our experimentally determined results to a set of well-defined literature values to gain a measure of the accuracy of our results. We measured the radius of CoRoT-1b to be $1.682 \times R_{\mathrm{Jup}}$. Compared to the literature value of $1.715 \pm 0.03 \times R_{\mathrm{Jup}}$ (Bonomo 2017), this result lies within two standard uncertainties of the literature. Our value is slightly lower than the literature, indicating there was less flux detected in this experiment. This aligns with the possible uncertainties arising from limb darkening and orbit inclination explored previously. We measured the mass of CoRoT-1b to be $1.41 \times M_{\mathrm{Jup}}$. Comparing this to the literature value of $1.23 \pm 0.1 \times M_{\mathrm{Jup}}$ (Bonomo 2017), our result is also within two standard uncertainties of the literature.

Since both results are within two standard uncertainties of the literature, we can infer that our results are highly accurate.

As part of a wider exploration of the astronomy and astrophysics community, let's explore an interesting extension of the transit method in determining the atmospheres of distant exoplanets. By examining the chemical composition of atmospheres of distant exoplanets we can gain insight into the planet's surface conditions, physical processes, and possibly even assess its ability to support life. Spectroscopic investigations of atmospheres of super-earths and related exoplanets holds out the best prospect of learning about these alien worlds (Tennyson 2017). Interestingly, the NASA Exoplanets Archive lists CoRoT-7b in the radius and mass-range of a rocky planet.

The transit spectroscopy method is performed by taking the spectrum of the light from the host star before, during, and after a transit (Aggarwal 2023). We can then compare them and infer the absorption features of the planet's atmosphere. Some common molecules and particles we can detect are water vapour, methane, and carbon dioxide. We generally refer to exoplanets with radii and mass similar to Earth with atmospheres composed of water, hydrogen, or helium as super-Earths. This field is extremely important to the ongoing search for life in the universe outside of Earth. Transmission spectroscopy has revolutionised our understanding of exoplanet atmospheres (Aggarwal 2017). This field has been driven further forward with the launch of the James Webb Space Telescope, providing further insight into the composition of these distance atmospheres.

4 Conclusion

This experiment thoroughly demonstrates the use of differential aperture photometry to detect flux drops across distant stars which correspond to transits of their exoplanets. We compared the flux of CoRoT-1b before and after a suspected transit, and then determined the characteristics of the exoplanet

CoRoT-1b during this transit. The results in this report lie within two standard uncertainties of the literature (Bonomo 2017). We found the radius of CoRoT-1b to be $1.682 \times R_{\mathrm{Jup}}$, and the mass to be $1.41 \times M_{\mathrm{Jup}}$. We also explored some significant areas of uncertainty in measuring flux differences that could have affected whether or not we detected a transit occuring, and also our radius and mass results. These involved limb darkening and the inclination of the orbit of the exoplanet. In future experiments, it would be beneficial to explore how carrying the standard error we found in the flux ratios through the radius and mass calculations would yield a more telling result. We could determine an uncertainty in our recorded results that could be used to suggest how precise our results are. Since our literature values have listed uncertainties, we were able to make the inference that our results were accurate, but we don't have a well-rounded measure of the precision of the results. We also explored the wider implications of differential aperture photometry beyond capturing the transit of a single exoplanet, and the exciting areas of study that are being pursued in this area.

5 References

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[3] Aliased 'Jovian Explorer' in the article

6 Appendices

6.1 Appendix 1

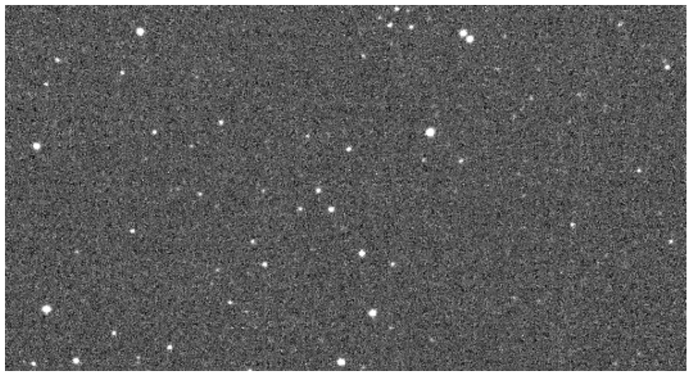

Figure 2: First image taken of CoRoT-1 on 21/03/2018 at 9:40:33 UTC.

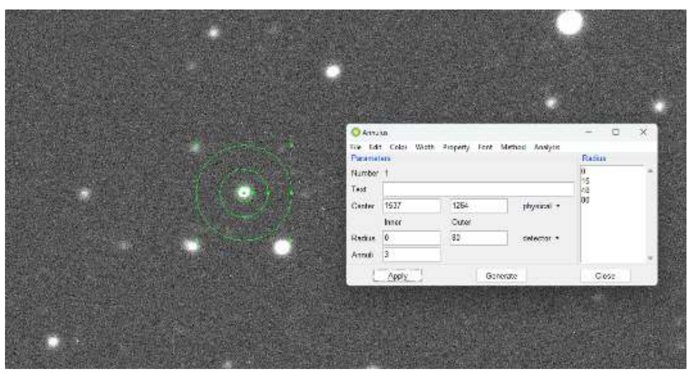

Figure 3: First image taken of CoRoT-1 with an annulus over the target star.

6.2 Appendix 2

		CoRoT-1				Reference Star				
Image Name	Image Number	Total inner radius sum	Background flux per pixel	Number of pixels	Flux of star	Total inner radius sum	Background flux per pixel	Number of pixels	Flux of star	Flux Ratio
13190	1	310510.38	268.393	709	120219.743	483515.7	267.405	709	293925.555	0.409014259
13191	2	313597.93	268.724	709	123072.614	488637.72	269.206	709	297770.666	0.413313426
13192	3	310949.99	267.565	709	121246.405	486107.29	266.645	709	297055.985	0.408160115
13193	4	307631.54	268.163	709	117503.973	474733.81	267.492	709	285081.982	0.412176077
13194	5	304451.29	267.503	709	114791.663	470216.66	266.924	709	280967.544	0.408558445
13195	6	301763.29	265.718	709	113369.228	457690.99	265.46	709	269479.85	0.420696494
13230	1	305267.56	267.639	709	115511.509	479256.2	266.968	709	289977.888	0.398345921
13231	2	306926.63	267.512	709	117260.622	475717.83	267.556	709	286019.206	0.409974641
13232	3	309482.8	268.01	709	119463.71	484680.38	267.584	709	294963.324	0.405012082
13233	4	301496.96	269.92	709	110123.68	462137.41	268.992	709	271422.082	0.405728521
13234	5	305148.62	269.161	709	114313.471	474600.73	268.868	709	283973.318	0.402550042
13235	6	302708.53	270.683	709	110794.283	467610.79	270.271	709	275988.651	0.401445069

Figure 4: Individual flux counts over regions for CoRoT-1 and reference star. Simplified down in Table 1.

6.3 Appendix 3

Main-Sequence Stars (Luminosity Class V)									
Sp. Type	T_e (K)	$L/L_\odot$	$R/R_\odot$	$M/M_\odot$	$M_{\rm bol}$	BC	M_V	$U-B$	$B-V$
O5	42000	499000	13.4	60	−9.51	−4.40	−5.1	−1.19	−0.33
O6	39500	324000	12.2	37	−9.04	−3.93	−5.1	−1.17	−0.33
O7	37500	216000	11.0	—	−8.60	−3.68	−4.9	−1.15	−0.32
O8	35800	147000	10.0	23	−8.18	−3.54	−4.6	−1.14	−0.32
B0	30000	32500	6.7	17.5	−6.54	−3.16	−3.4	−1.08	−0.30
B1	25400	9950	5.2	—	−5.26	−2.70	−2.6	−0.95	−0.26
B2	20900	2920	4.1	—	−3.92	−2.35	−1.6	−0.84	−0.24
B3	18800	1580	3.8	7.6	−3.26	−1.94	−1.3	−0.71	−0.20
B5	15200	480	3.2	5.9	−1.96	−1.46	−0.5	−0.58	−0.17
B6	13700	272	2.9	—	−1.35	−1.21	−0.1	−0.50	−0.15
B7	12500	160	2.7	—	−0.77	−1.02	+0.3	−0.43	−0.13
B8	11400	96.7	2.5	3.8	−0.22	−0.80	+0.6	−0.34	−0.11
B9	10500	60.7	2.3	—	+0.28	−0.51	+0.8	−0.20	−0.07
A0	9800	39.4	2.2	2.9	+0.75	−0.30	+1.1	−0.02	−0.02
A1	9400	30.3	2.1	—	+1.04	−0.23	+1.3	+0.02	+0.01
A2	9020	23.6	2.0	—	+1.31	−0.20	+1.5	+0.05	+0.05
A5	8190	12.3	1.8	2.0	+2.02	−0.15	+2.2	+0.10	+0.15
A8	7600	7.13	1.5	—	+2.61	−0.10	+2.7	+0.09	+0.25
F0	7300	5.21	1.4	1.6	+2.95	−0.09	+3.0	+0.03	+0.30
F2	7050	3.89	1.3	—	+3.27	−0.11	+3.4	+0.00	+0.35
F5	6650	2.56	1.2	1.4	+3.72	−0.14	+3.9	−0.02	+0.44
F8	6250	1.68	1.1	—	+4.18	−0.16	+4.3	+0.02	+0.52

Figure 5: Literature table of data and characteristics of main sequence stars according to stellar classification.

6.4 Appendix 4

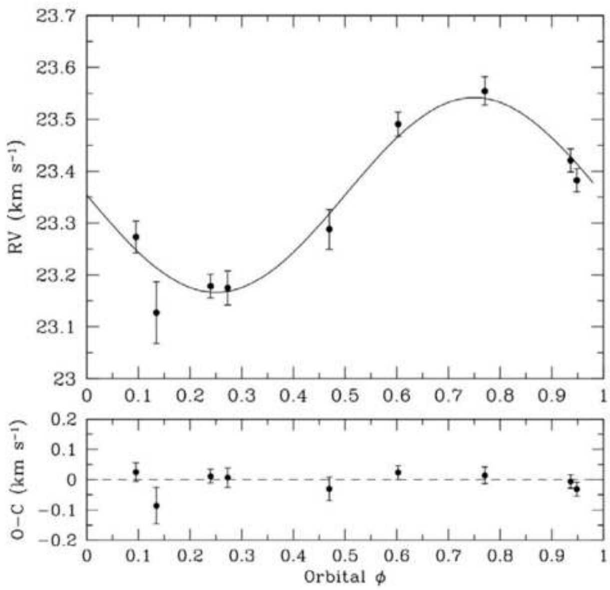

Figure 6: Radial velocity diagram of CoRoT-1, showing the radial velocity against orbital phase (top) and residuals from sinusoidal best fit (bottom).